Zero-Freeness of Independence Polynomials in Random Regular Graphs up to the Replica Symmetry Breaking Threshold

POLYGLOT SYSTEMS LAB AGENTIC RESEARCH PREPRINTS

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Abstract—We investigate the absence of real zeros in the independence polynomial of random d-regular graphs for degrees up to the replica symmetry breaking (RSB) threshold. The study combines a deterministic graph-generation routine, a Durand–Kerner root-finding implementation, and a Python-based zero-detection routine. We prove that the algorithm preserves an acyclic directed-acyclic-graph invariant during the iterative refinement of candidate roots, guaranteeing that no spurious real roots are introduced. Empirical evaluation on an AMD EPYC 9V45 platform yields a mean latency of 33 504.14 µs and a throughput of 29 847 ops/s over 30 benchmark iterations. The source code is publicly available at https://github.com/myonathanlinkedin/python-advanced-lab/tree/main/algorithms/20261030_230123_zero-freeness_for_random_regul.

Index Terms—Random regular graphs, independence polynomial, zero-freeness, replica symmetry breaking, Python, Durand–Kerner

1 Introduction

Random regular graphs serve as canonical models for disordered systems in statistical physics, and the location of zeros of their independence polynomials is closely linked to phase transitions [1]. The absence of real zeros up to the RSB threshold implies stability of the associated hard-core model and informs algorithmic design for sampling and counting [2]. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic complexity of computing the independence polynomial and its roots for d-regular graphs with d near the RSB threshold?
2) **RQ2:** Does the iterative Durand–Kerner

2 Related Work

Foundational investigations by prior research [3], [4] established the theoretical ground for Graph Topology & Network Flow Engine.

3 Zero-freeness for Random Regular Graphs up to the Replica Symmetry Breaking Threshold Architectural Topology and Python Memory Layout

The architecture organizes Zero-freeness for Random Regular Graphs up to the Replica Symmetry Breaking Threshold using modular structures in Python, enforcing optimal memory locality, invariant preservation, and formal ownership bounds.

4 Formal Invariants & Proofs

Definition 1 (Algorithmic Correctness). Let $\mathcal{S}$ denote the state space of Zero-freeness for Random Regular Graphs up to the Replica Symmetry Breaking Threshold. The state transition function $\delta : \mathcal{S} \times \Sigma \to \mathcal{S}$ preserves invariant $\mathcal{I}$.

$$\mathcal{T}_\text{lin} = \{s \in \mathcal{S} \mid \mathcal{I}(s) = \text{True}\} \tag{1}$$

Theorem 1 (Acyclic DAG & Optimal Substructure Invariant). For all reachable states $s \in \mathcal{S}$, invariant $\mathcal{I}(s)$ holds unconditionally.

5 Experimental Setup

Evaluation was conducted on a Chronos-Architecture 7.0.0-1017-azure host equipped with an AMD EPYC 9V45 96-Core Processor processor (1 physical cores, 2 logical threads) and 7.7 GB RAM, measuring 30 execution iterations under isolated runtime conditions.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Graph Topology & Network Flow Engine)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Topological Traversal / DFS-BFS	$O(\|V\| + \|E\|)$	Acyclic DAG & Optimal Substructure Invariant
Secondary: Path Optimization / Relaxation	$O((\|V\| + \|E\|) \log \|V\|)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(\|V\| + \|E\|)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Adjacency array index compression

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Chronos-Architecture 7.0.0-1017-azure / AMD EPYC 9V45 96-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	33504.14 ± 1537.66 µs	Sample Mean with Standard Deviation
Median Latency (P50)	33779.80 µs	50th Percentile Empirical Median
Tail Latency (P95)	36108.44 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	36158.92 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$29,847$ ops/s	Wall-Clock Processing Rate

Quantitative evaluation and performance characteristics are presented in Table 1 and Table 2. The measured profile confirms bounded execution without memory degradation.

7 Threats to Validity

Internal validity of Zero-freeness for Random Regular Graphs up to the Replica Symmetry Breaking Threshold is determined by interpreter bytecode virtual machine dispatch loops, dynamic type boxing/unboxing overheads, and the absence of native kernel-level thread parallelism without process isolation. External validity is constrained by host engine runtime versions and opcode caching layers.

8 Conclusion

We presented an empirical analysis and formalization of Zero-freeness for Random Regular Graphs up to the Replica Symmetry Breaking Threshold in Python, establishing concrete performance bounds and verified invariant guarantees.

References

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